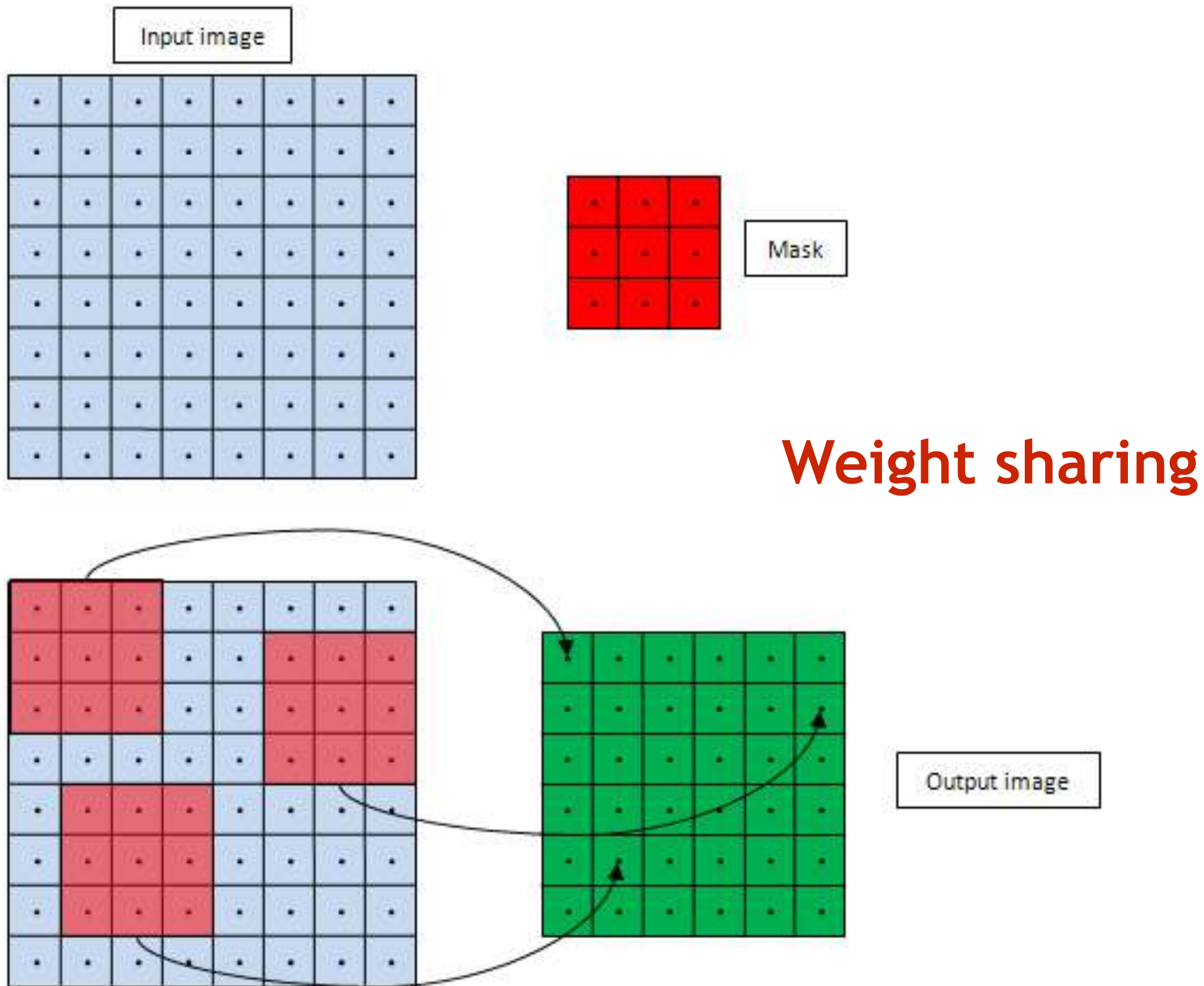


E9 205 Machine Learning for Signal Processing

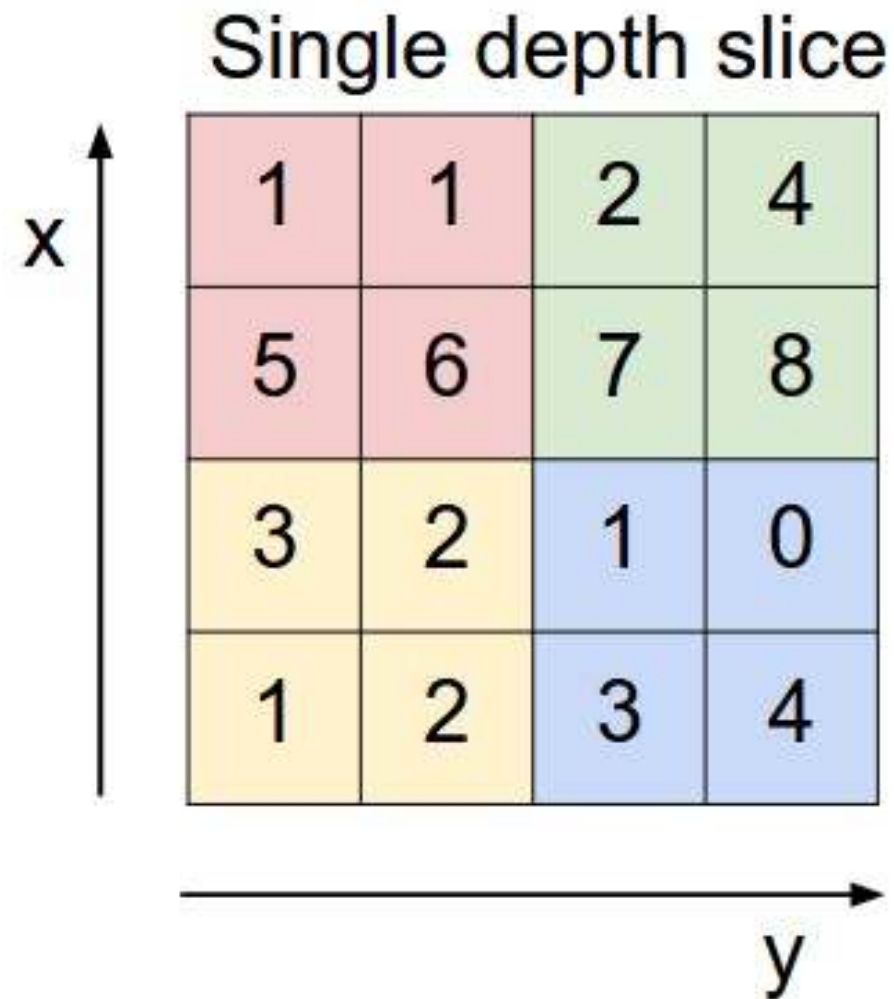
Convolutional Neural Networks

2-11-2018

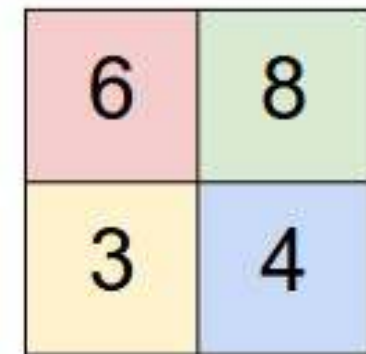
Other Architectures - Convolution Operation



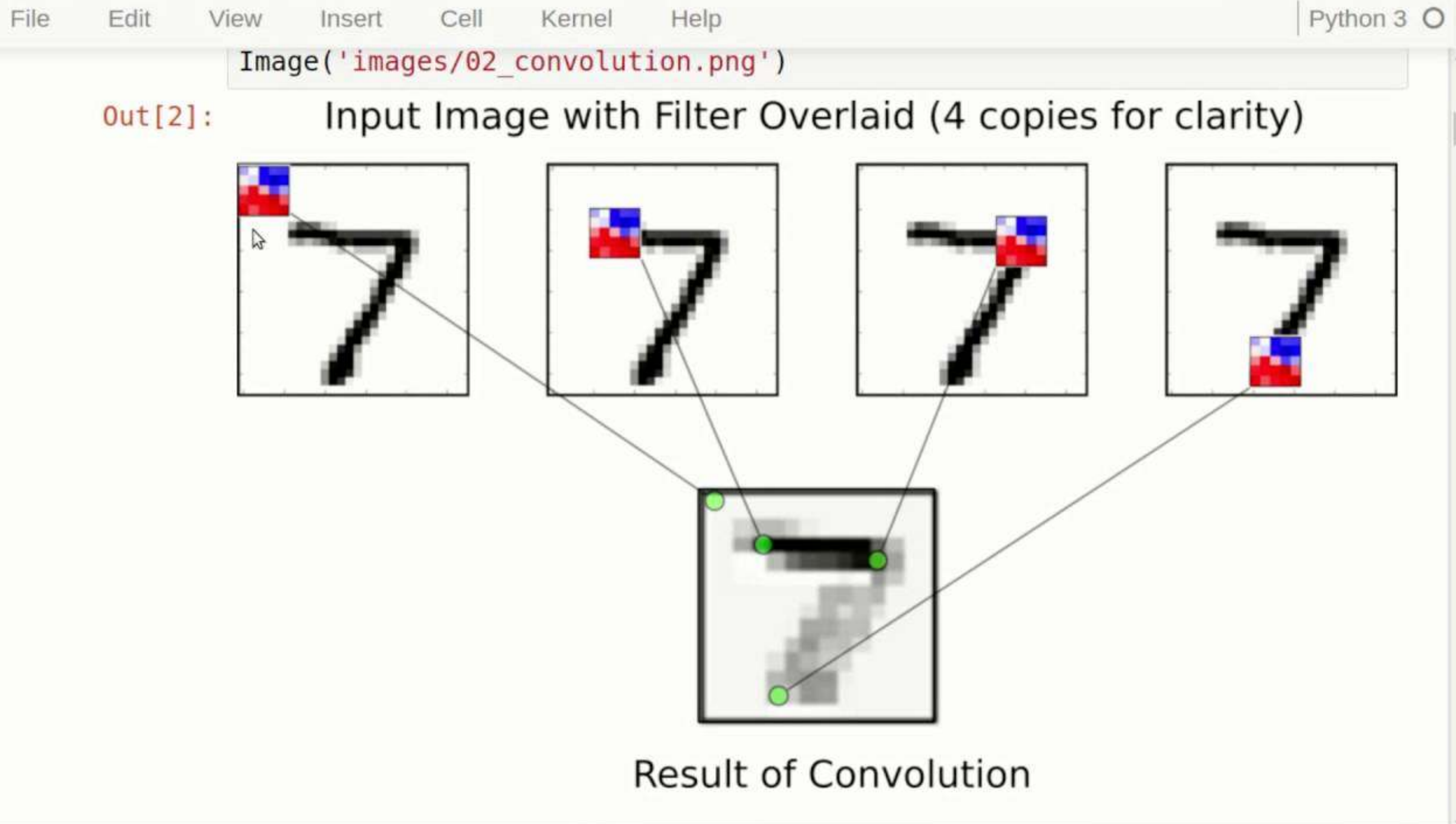
Max Pooling Operation



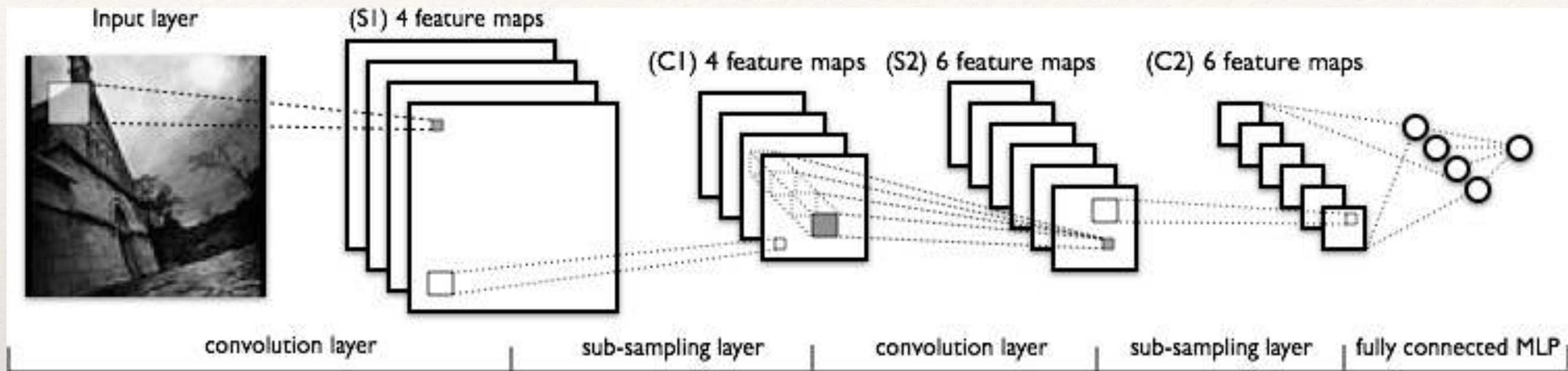
max pool with 2x2 filters
and stride 2



Convolution Operation (Images)

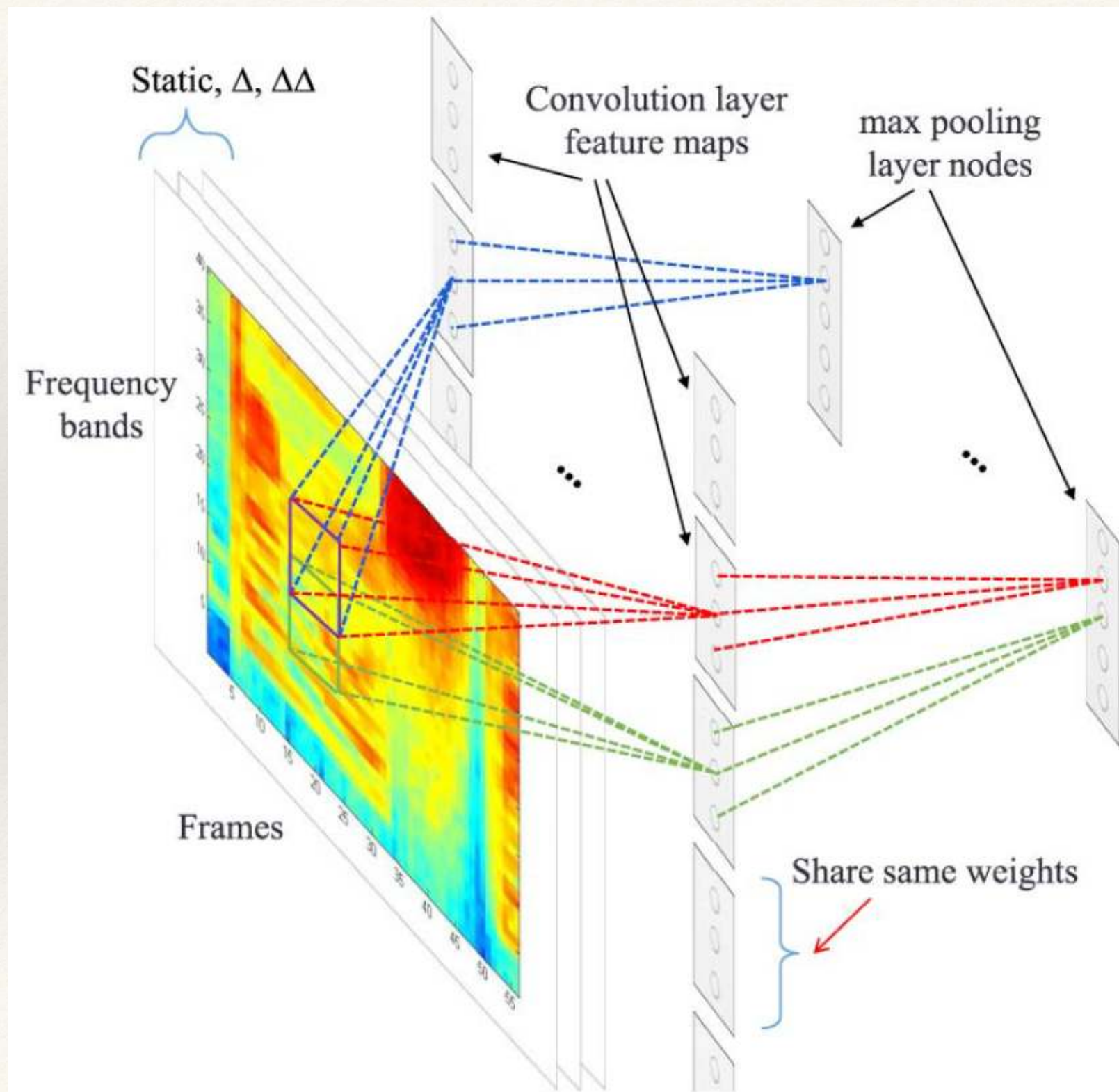


Convolutional Neural Networks



- Multiple levels of filtering and subsampling operations.
- Feature maps are generated at every layer.

CNNs for Speech and Audio



Representation Learning in CNNs

■ It's **deep** if it has **more than one stage** of **non-linear feature**

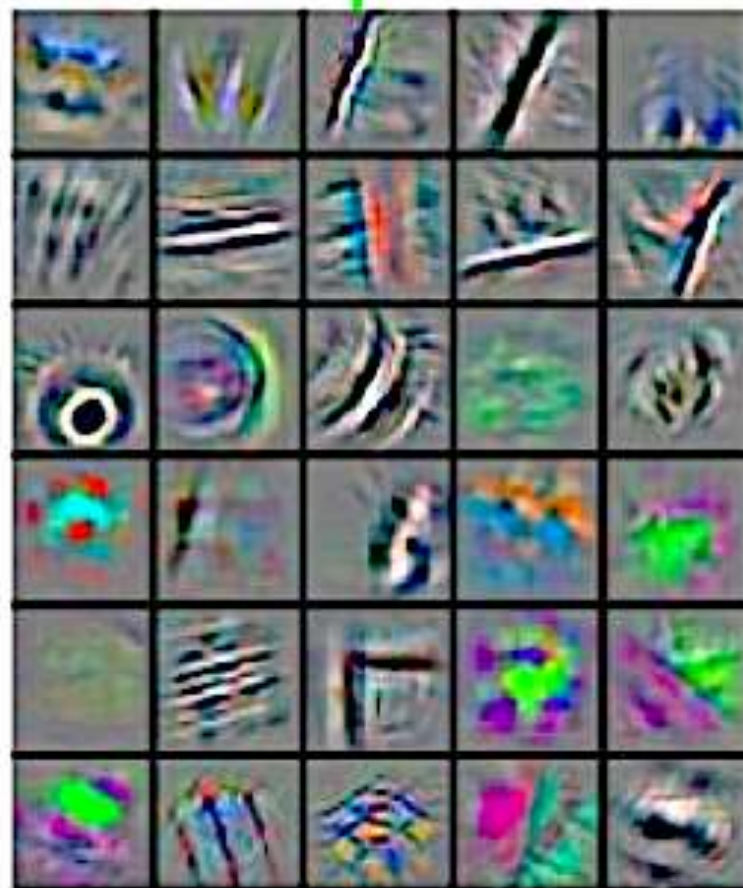


Low-Level
Feature

Mid-Level
Feature

High-Level
Feature

Trainable
Classifier



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Representation Learning in CNNs

[Zeiler, 2014]

